

Exercise: Part 1 Network Security Lab

Download Instructions

Download IP Fire [here](#)

Configuring Your VM

Using [VirtualBox](#) configure a virtual machine. (Note, if you have an additional device with 2 network cards you can also configure this way in place of a VM. 1 will need to be red connecting to the internet and 1 will be green connecting back to your management device/client).

When entering in the root password, you will not see what you are typing in, but the password is being entered you just won't see it or it won't be masked, just know it is entered when typing.

Please use the below settings to configure:

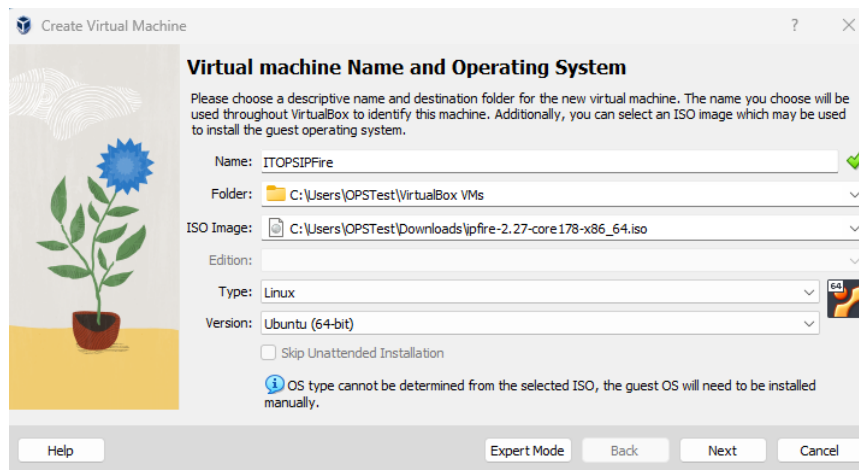
25GB Storage

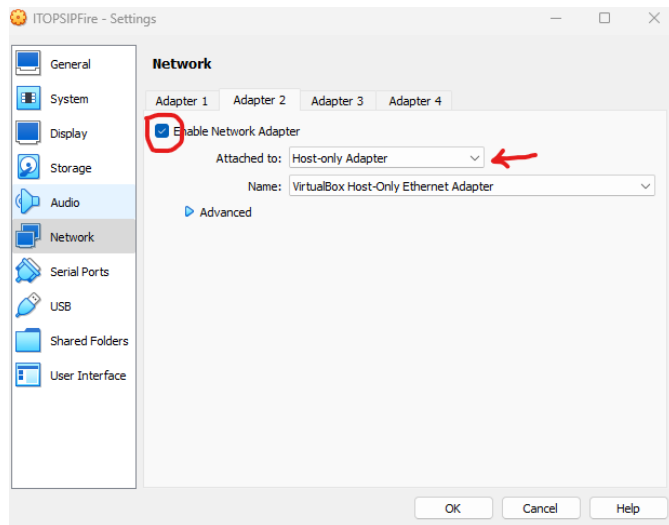
1 CPU Core

2GB Ram

(Note: The more solutions you add to IP Fire and the more users, the more resources will be required.)

On the virtual machine you will need to create a second virtual network interface card. On the first nic configure it to use NAT. On the second card you will use host only. Please make note of the MAC Addresses of each.





Helpful Hint!

To navigate the configuration interface you will need to become very familiar with the Tab Key Space Key and Enter Key on your keyboard. Your mouse will not work.

Start your Installation

Review your license information.

Question response 1: Are there any considerations that stand out to you? Is there any red flags you would need to provide to your manager as part of this review of the software firewall?

Follow the prompts to continue the installation

Reboot, follow the prompts again to continue the basic configurations

Continuing Configurations

Name your machine hostname based on the following example SECFFIRSTNAMELASTINITIAL as an example a student with the name of Jon T. Test would be SECFJONT

Leave the domain as localdomain.

Create

a strong root password.

Please remember if you lose this password you will have to start over.

Create a strong admin password.

Please remember if you lose this password you will have to start over.



Network configuration

First, on your host device do an IPConfig /all on the workstation host the VM to find your Virtual adapter address range. Note this is NOT your regular network adapter IP Address. If using VirtualBox it should be starting with a 192.168

New, please select:

Red+Green

Under Drivers and card assignments:

On a virtual machine you will need to create a second virtual network interface card while the VM is off if you forgot that step in the initial configuration section.

Assign the **Red** network to the NAT adapter (Check your MAC address to ensure they match, this will break your configuration and give you significant headaches if you misconfigure at this point)

Assign the **Green** network to the Host Only adapter (Again, check your MAC!)

If you were using a physical machine with two nics you would need to put the red zone to the outside/front facing and green to your trusted zone.

Under address settings you will need to set the address type for the red zone. Use DHCP for this while on campus. If you had a static configuration from your ISP you may configure it here but for the purpose of this activity you will use DHCP.

You will need to set an ip address for the green interface as well.

Leave the DHCP configuration unchecked for the moment.

Select done and allow for the device to process a reboot.

Access The WebUI

Access the Webui by using your web browser of choice on the host PC and typing in the full IP address followed by :444.

As an example it should look like: `HTTPS://192.168.66.66:444`

Response Question 2: What is the relevance to the number following the IP Address? If you saw another device connecting using that port to your network, what could it mean?

You will need to enter the username and password you created when you configured the device initially. Use the admin account to access the WebUI.

Take a screenshot of your device configuration/call your instructor over to confirm your device configuration. Hand in your two question responses at the end of checkpoint 1 and include a copy of the screenshot.



Part 2 Configurations

Explore the WebUI

Now that we've setup our software firewall, take a few minutes to explore the environment and available settings/configurations.

Response Question 3: Was there service or configuration that was unexpected or surprised you? Why or why not?

Check for Hardware Vulnerabilities

Go to System, and Hardware Vulnerabilities. Check for any vulnerabilities that the system detects. Take a screenshot and include this in your response file including your device host name we configured in part 1.

Enable Web Proxy

We're going to go to the NETWORK tab in the WebUI and click on the Web Proxy

You're going to need to enable on green and enable transparent on green.

Make note of the port number and settings, you will need this to configure your device to test the proxy. (In a enterprise environment, you would likely hand out the proxy configuration settings through group policy or dhcp. Since we're just testing here, we will use the manual configuration)

Enable the URL filter.

Enable all 3 log settings.

Response Question 4: What benefits might the update accelerator provide? If you're unsure of the function check back on the IPFire wiki.

Save your settings and move on to the next step. Please ensure you double check your configuration as this could impact your ability to complete your next steps.

Response Question 5: What are some of the benefits of using a web proxy configuration? What are some of the risks/downsides?

Important** Ensure you save and restart the device service as found at the bottom of the page.

URL Filtering

Next we're going to go to the NETWORK tab in the WebUI and click on URL Filter

Configure blocked URLs

Check off all the boxes for URL filters

In the custom blacklist enter the below sites:

1. Kijiji.ca
2. facebook.com

3.reddit.com

4.youtube.com

Under the file extension blocking, select block executable files

Under block Page Settings

Check off the show category, show url, and show ip on block page. In the message 1 line, put your device name you used earlier – SECFJONT as the example.

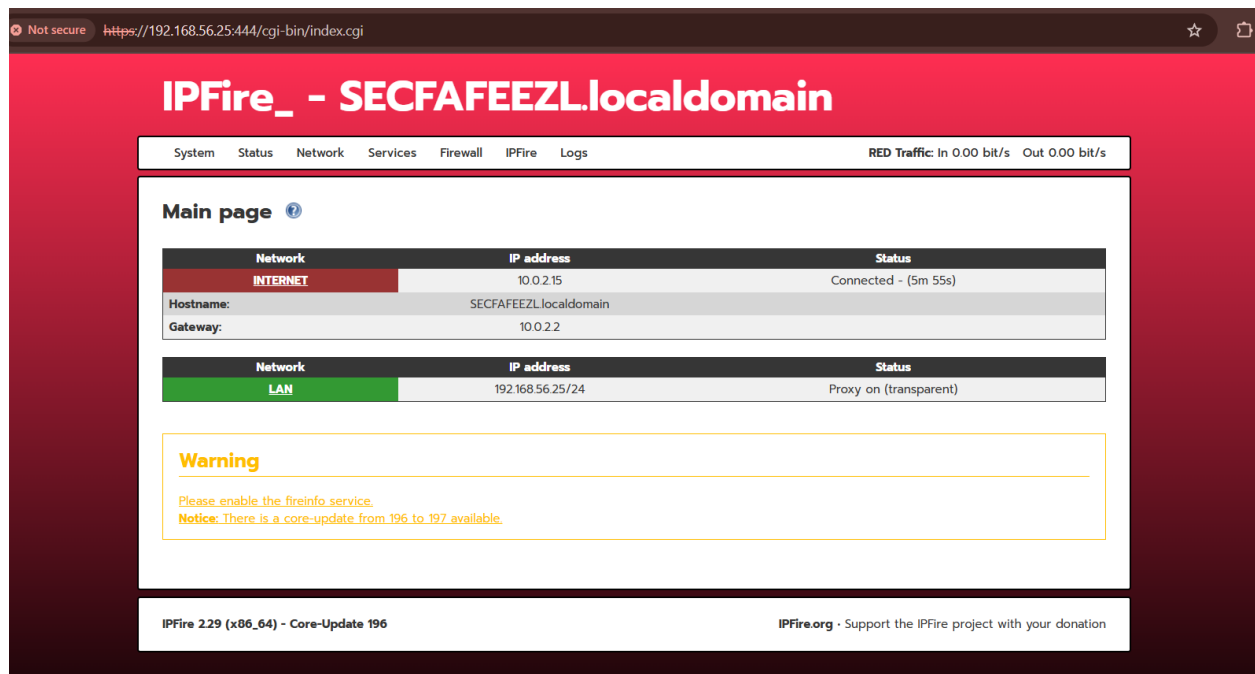
Take a screenshot of your completed configuration and include it in your response document.

Question 1: Are there any considerations that stand out to you? Is there any red flags you would need to provide to your manager as part of this review of the software firewall?

- IPFire is a well run project and it is strong in the areas that matter like controlling traffic and basic firewalling.
- While IPFire is very capable, it's an open-source firewall. That means it doesn't come with vendor support like commercial firewalls, instead the community provides frequent updates and patches which is a big strength.

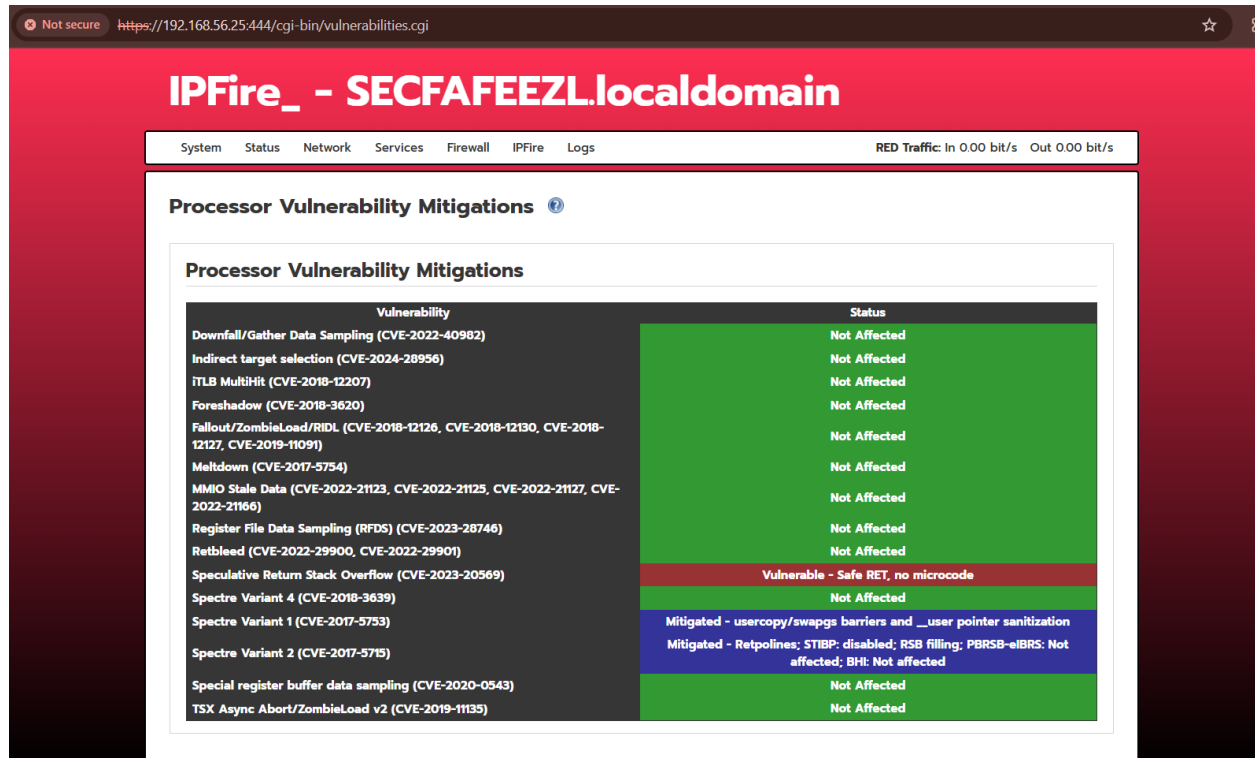
Question 2: What is the relevance to the number following the IP Address? If you saw another device connecting using that port to your network, what could it mean?

- The number :444 after the IP Address is the TCP port number. When we type in our web address then :444 it means we opened a browser session to the IPFire admin page on port 444. Port 444 is where IPFire management interface listens.
- If I saw another device connecting to that port, if it is expected it could mean somebody is trying to login to manage the firewall or if it is unexpected, it is most likely a security risk.



Question 3: Was there service or configuration that was unexpected or surprised you? Why or why not?

- I had to navigate using my keyboard and there was no mouse support, that was unexpected

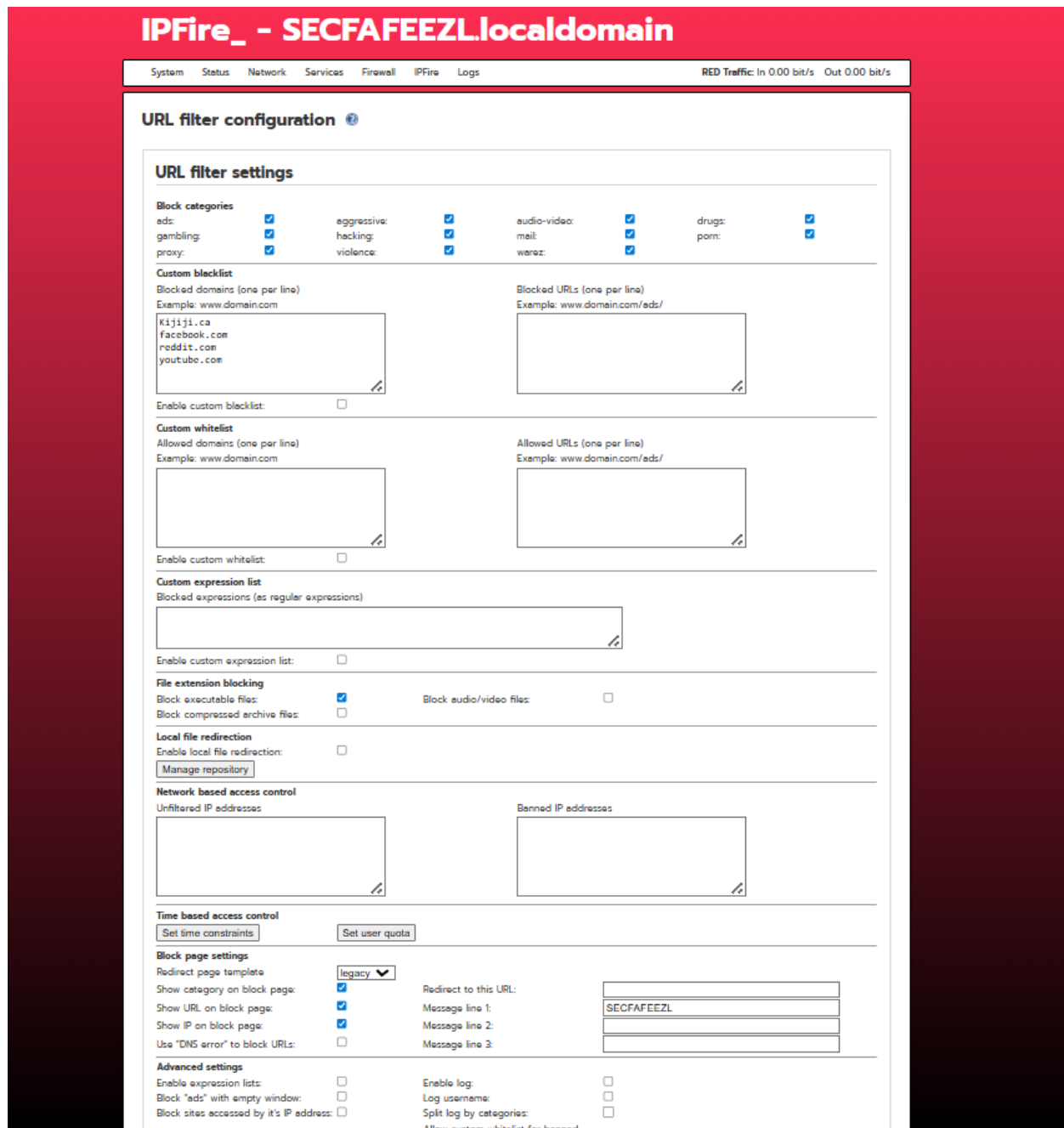


Question 4: What benefits might the update accelerator provide? If you're unsure of the function check back on the IPFire wiki.

- The Update Accelerator in IPFire is a local cache, basically when one computer downloads an update, other computers on the same network can grab the update locally. IPFire makes a copy of an update when it's downloaded through its firewall and stores it locally, when another computer tries to download the update from the internet through IPFire firewall it automatically hands that file over from its cache. It reduces stress on network by aiding download.

Question 5: What are some of the benefits of using a web proxy configuration? What are some of the risks/downsides?

- Benefits – web proxy sits between users and the internet. It helps control user traffic, and we can decide what website people can access or not. It also adds another layer of security since we can filter bad sites.
- Risk – it can sometimes block legitimate traffic or slow users down if it's not tuned well.



Question 6: Why might you start configuring your filtering by blacklisting all URLs and then building your whitelist? What are some of the impacts of this configuration?

- Starting with a blacklist-all approach means the firewall blocks everything by default and then we only allow access to sites we specifically trust through a whitelist.
- The impact is that employees will need to get used to it because they won't be able to access legitimate sites they could access before.

Question 7: What are some cases where an open-source firewall would be of benefit? What situations might it be better to use an off the shelf model?

- An open-source firewall like IPFire is a great fit in environments where budget is tight and technical staff are comfortable with hands-on configuration. For example, in a small business, an open-source firewall will provide strong protection at low cost, with frequent updates, and the ability to customize features as needed.
- In larger enterprises or regulated industries, an off-the-shelf commercial firewall is better because they come with vendor support.